NERVOUS SYSTEM
DIVISIONS CNS, PNS-1

LEARNING OBJECTIVES

At the end of the lecture the student should be able to;

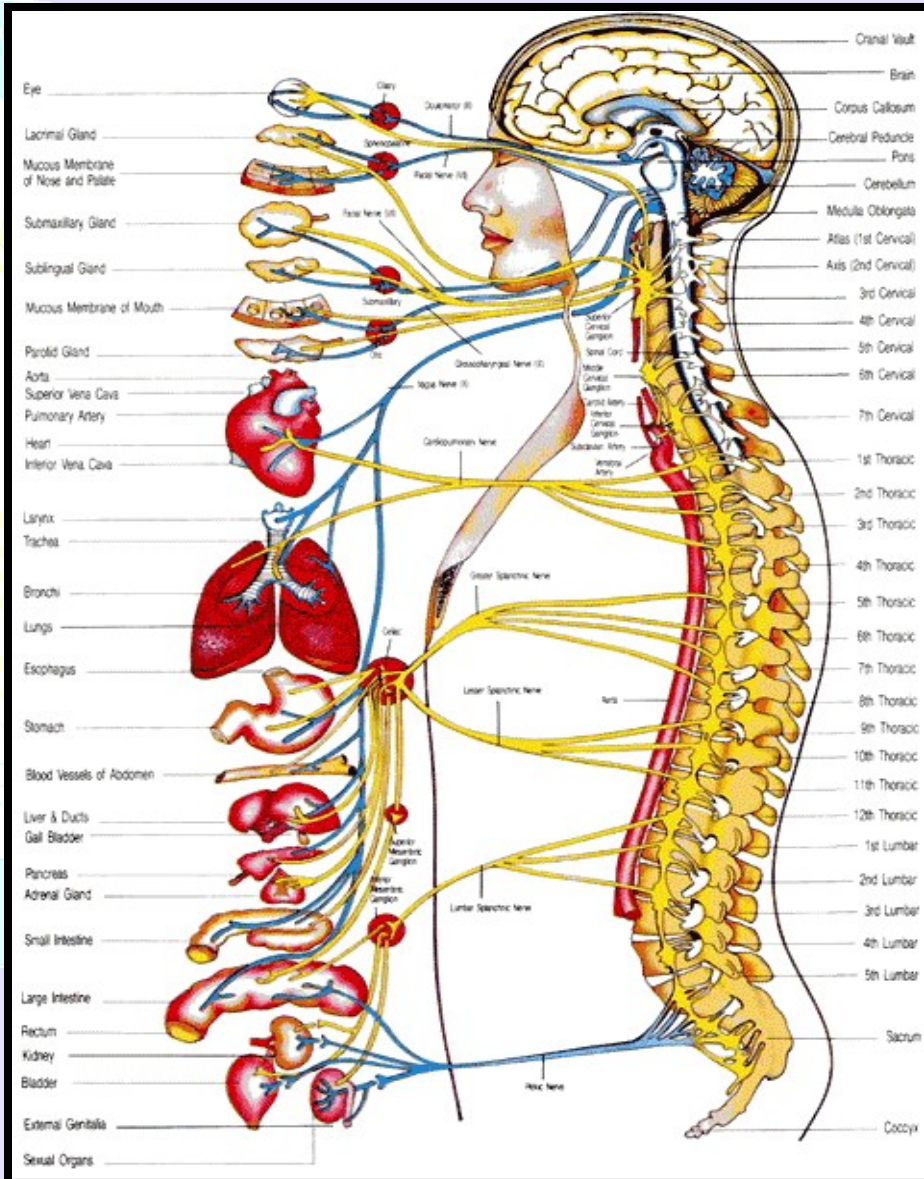
- Enlist the basic divisions of Nervous system
-
- Enlist and define the various components of CNS and PNS

- Describe the structure of Neuron

- Describe the processes of neurons

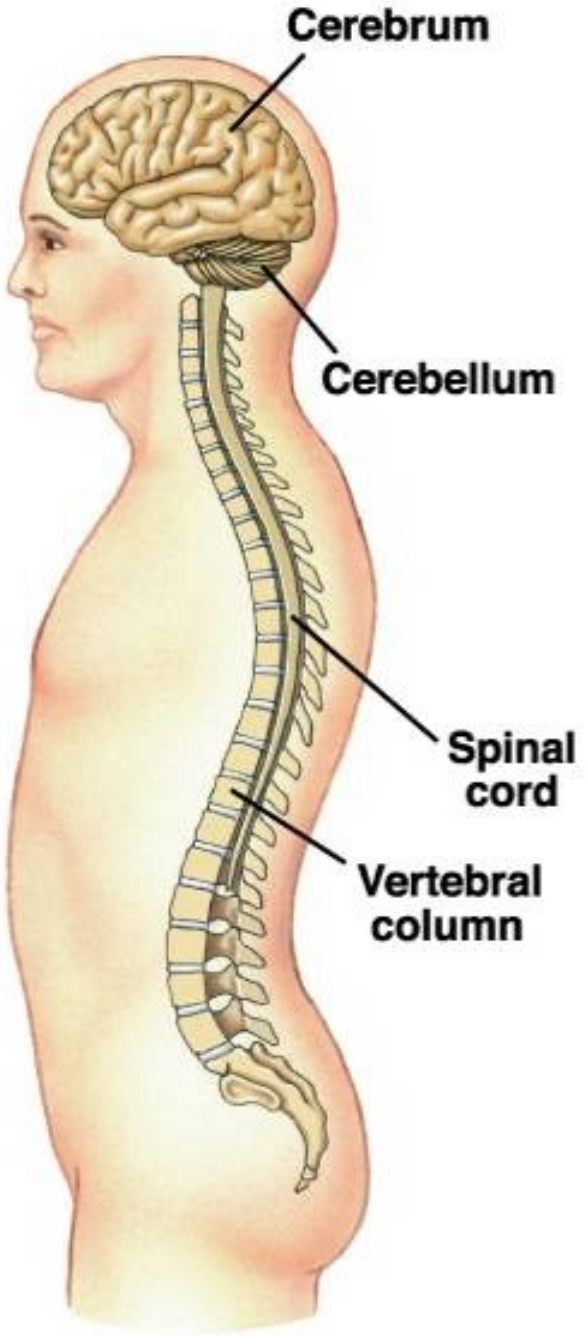
NERVOUS SYSTEM

- Highly organized system of human body.
- It is the organ system containing a network of specialized cells called neurons that coordinate the actions of an animal and transmit signals between different parts of its body.



ORGANIZATION OF THE NERVOUS SYSTEM

- Anatomical Divisions:
  1. **Central Nervous System**
     - The brain + the spinal cord →
       - The center of integration and control
  2. **Peripheral Nervous System**
     - The nervous system outside of the brain and spinal cord
     - Consists of:
       - 31 Spinal nerves
         - » Carry info to and from the spinal cord
       - 12 Cranial nerves
         - » Carry info to and from the brain



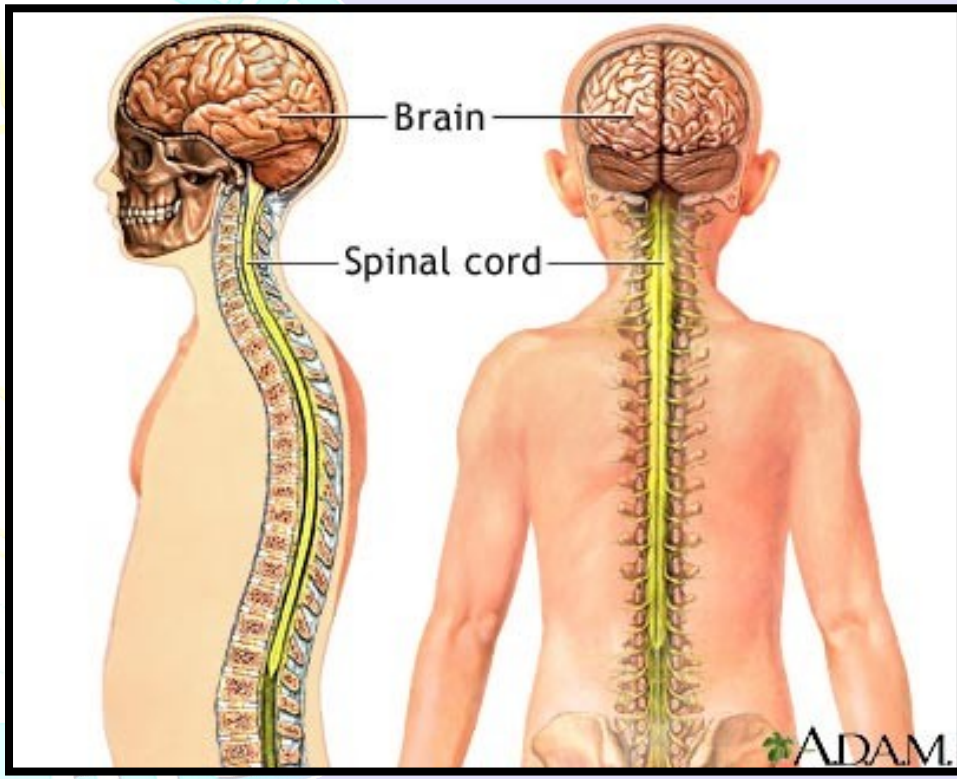
BASIC FUNCTIONS OF THE NERVOUS SYSTEM

1. **Sensation:**
   - Monitors changes/events occurring in and outside the body. Such changes are known as stimuli and the cells that monitor them are receptors.
2. **Integration:**
   - The parallel processing and interpretation of sensory information to determine the appropriate response
3. **Reaction/Response:**
   - Motor output.
   - response to information processed through stimulation of *effectors*
     - muscle contraction, glandular secretion

The activation of muscles or glands (typically via the release of neurotransmitters (NTs))

CENTRAL NERVOUS SYSTEM

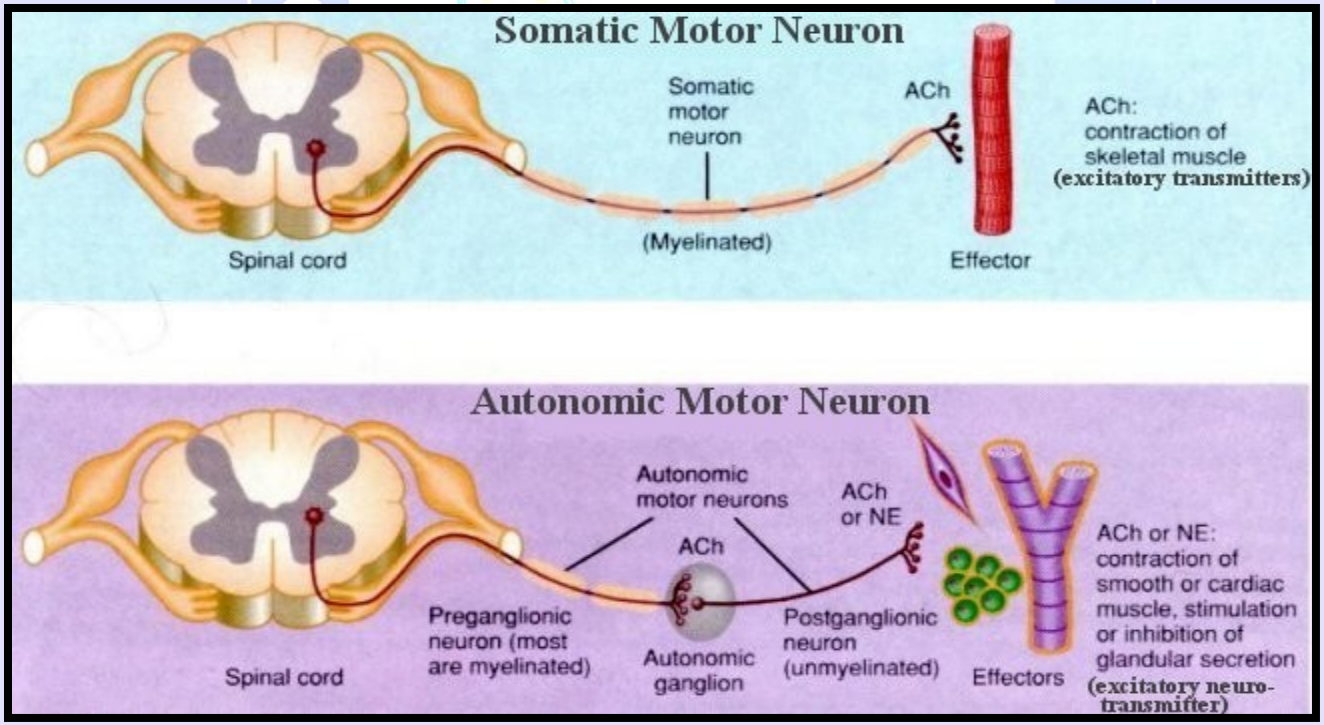
- Divided into two parts:
- ***Central Nervous system:***
  - **Consists of**
  - **Brain,** contained within the cranium
  - **Spinal cord,** lodged in the vertebral canal
  - Two portions are continuous with one another at the level of the upper border of the atlas vertebra.



Peripheral nervous system

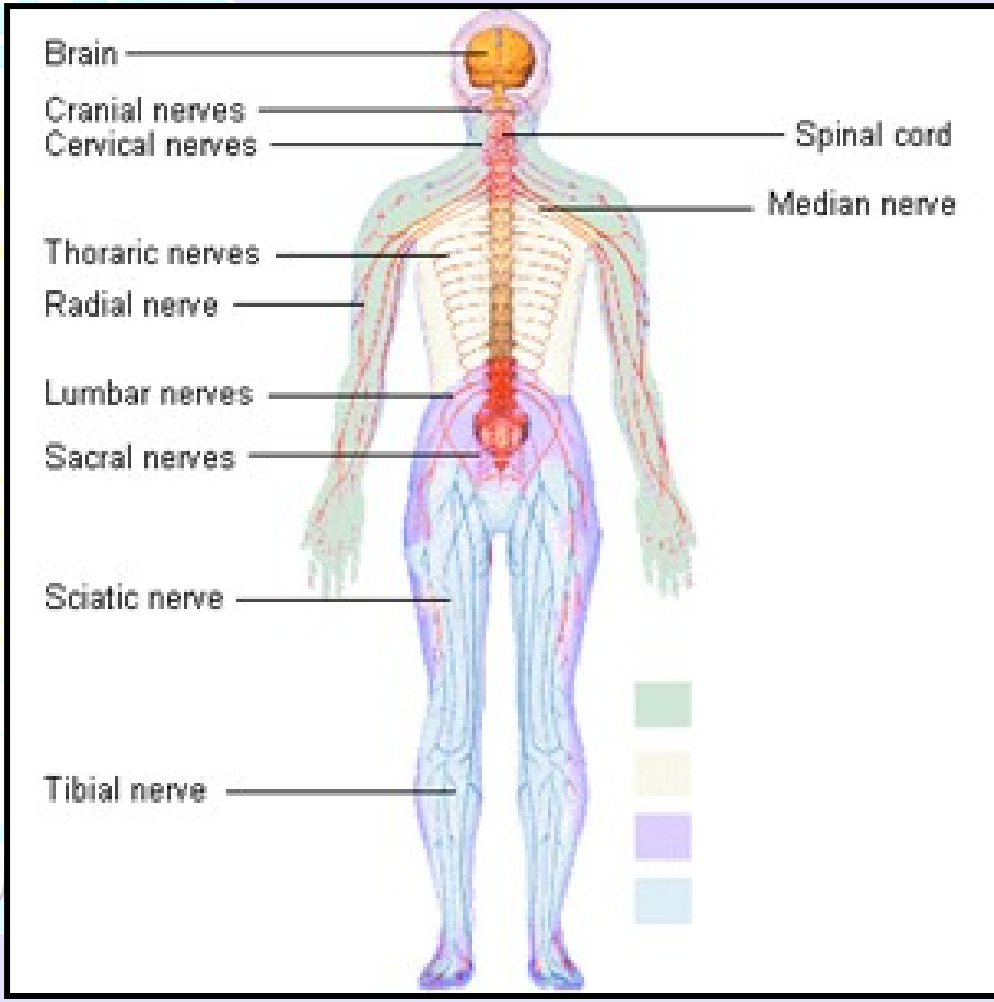
- ***Peripheral nevous system:***

- It is a series of nerves by which the central nervous system is connected with the various tissues of the body.
- PNS is not protected by the bone of spine and skull, or by the blood-brain barrier, leaving it exposed to mechanical injuries.
- Divided into:
  - Somatic nervous system
  - Autonomic nervous system



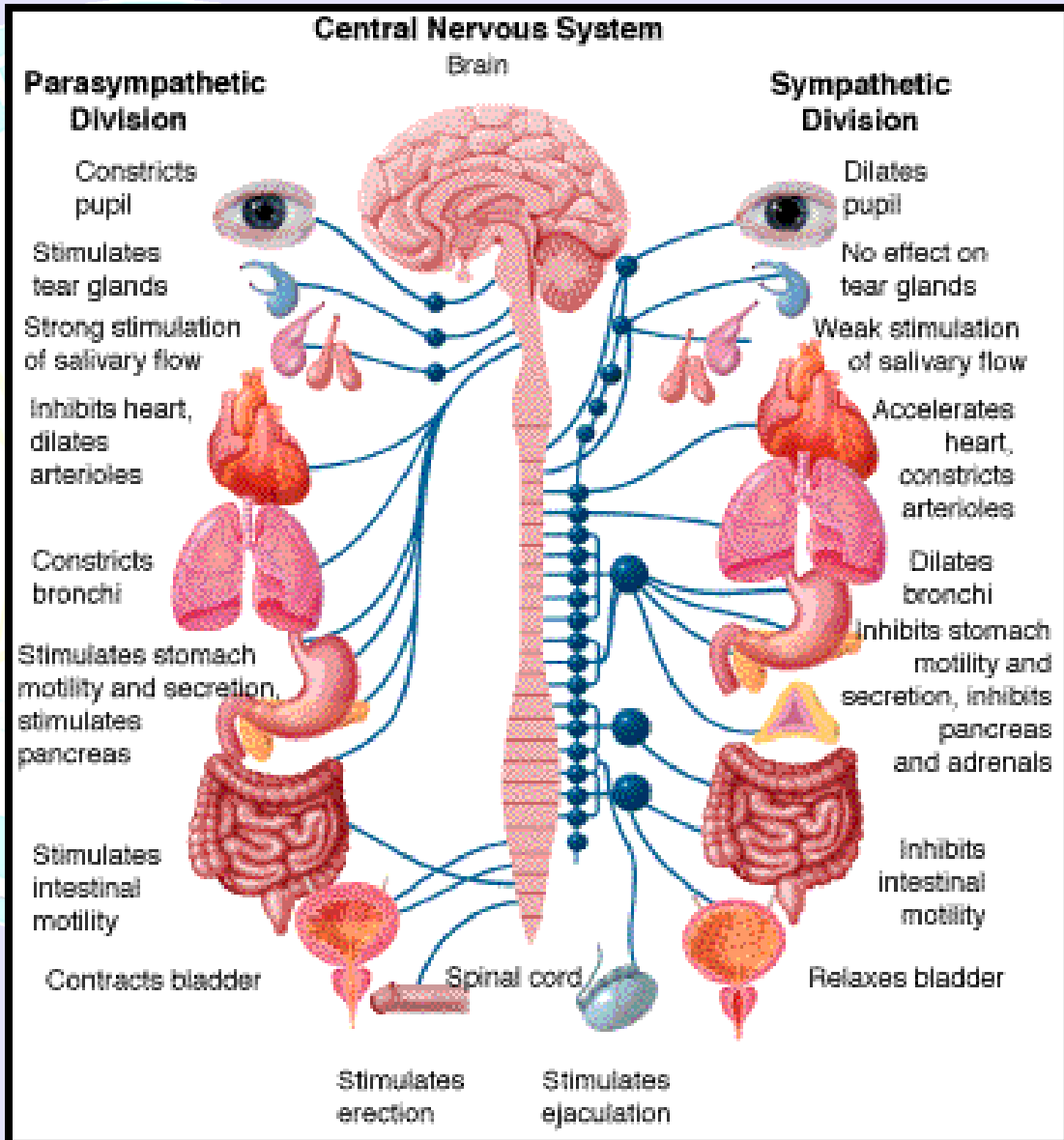
SOMATIC NERVOUS SYSTEM

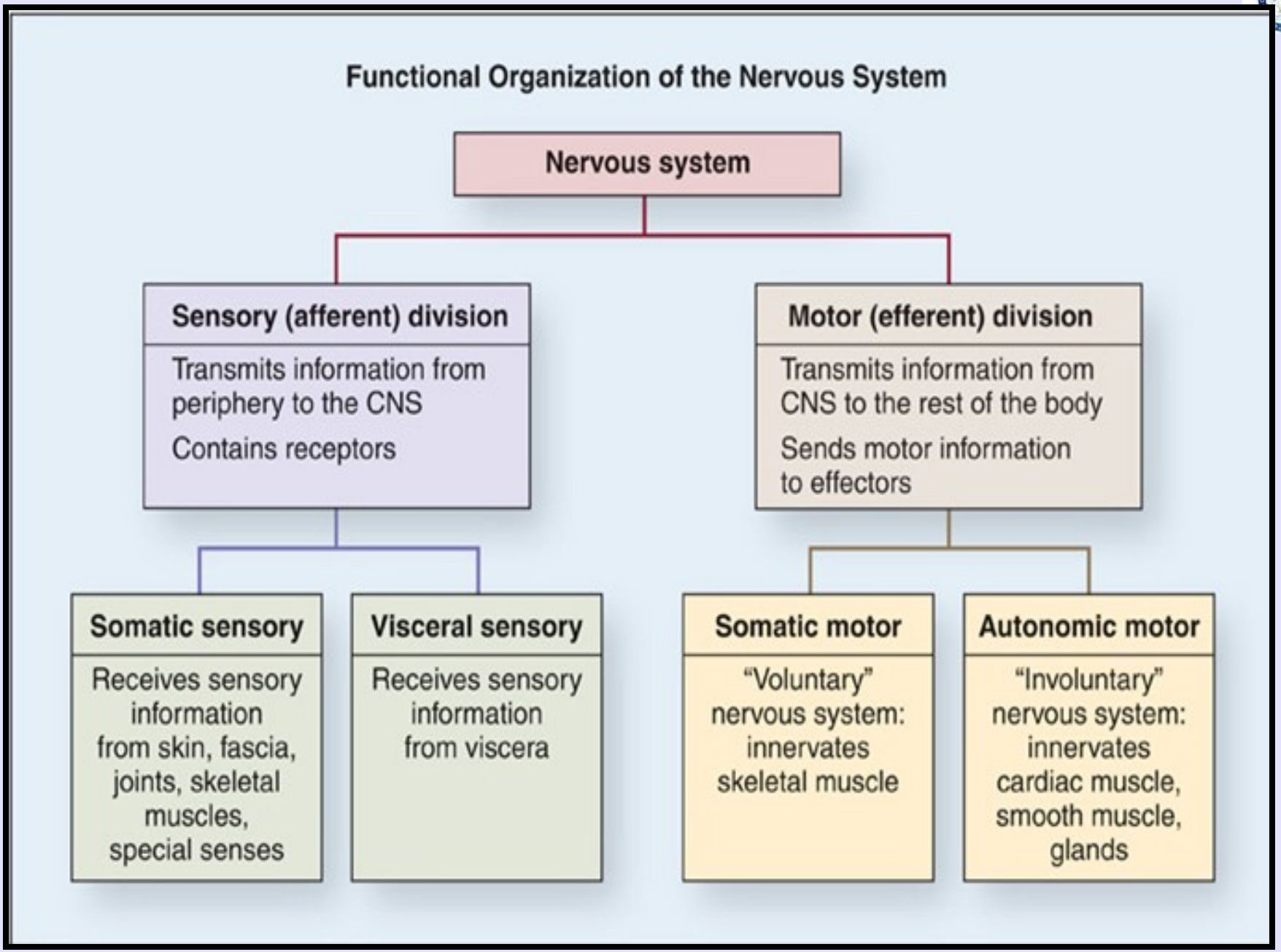
- Associated with the voluntary control of body movements via skeletal muscles, and sensory reception of external stimuli (e.g., touch, hearing, and sight)
- Cerebrospinal nerves are 43 in number on either side
- 12 **cranial,** attached to the brain
- 31 **spinal,** attached to the spinal cord.



AUTONOMIC NERVOUS SYSTEM

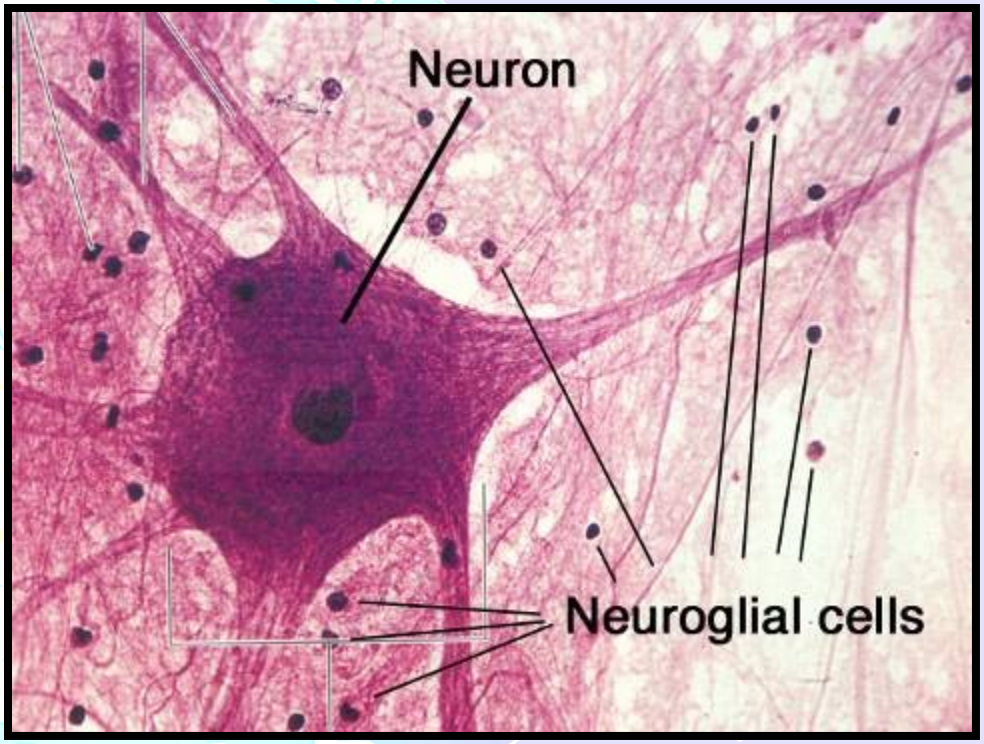
- The ANS affects heart rate, digestion, respiration rate, salivation, perspiration, diameter of the pupils, micturition (urination), and sexual arousal.
- Whereas most of its actions are involuntary, some, such as breathing, work in tandem with the conscious mind.




Functional Organization of the Nervous System
Nervous system
Sensory (afferent) division
Transmits information from periphery to the CNS
Contains receptors
Motor (efferent) division
Transmits information from CNS to the rest of the body
Sends motor information to effectors
Somatic sensory
Receives sensory information from skin, fascia, joints, skeletal muscles, special senses
Visceral sensory
Receives sensory information from viscera
Somatic motor
"Voluntary" nervous system: innervates skeletal muscle
Autonomic motor
"Involuntary" nervous system: innervates cardiac muscle, smooth muscle, glands

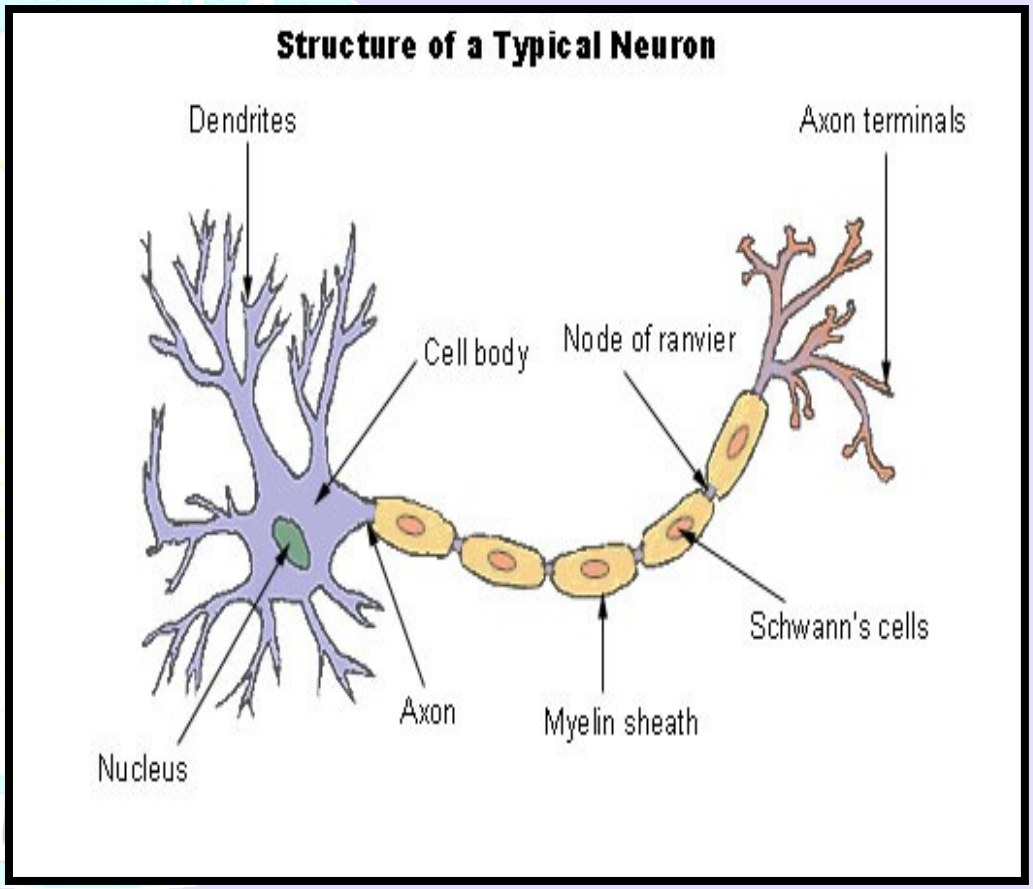
NERVOUS TISSUE

- Nervous Tissue comprises of:
  - Neurons
  - Supporting cells or Neuroglia.



NEURON

- The structural and functional unit of the nervous system.
- Neurons are highly specialized cells that easily conduct electrical impulses and are easily excited to produce them (a property of a cell known as irritability).

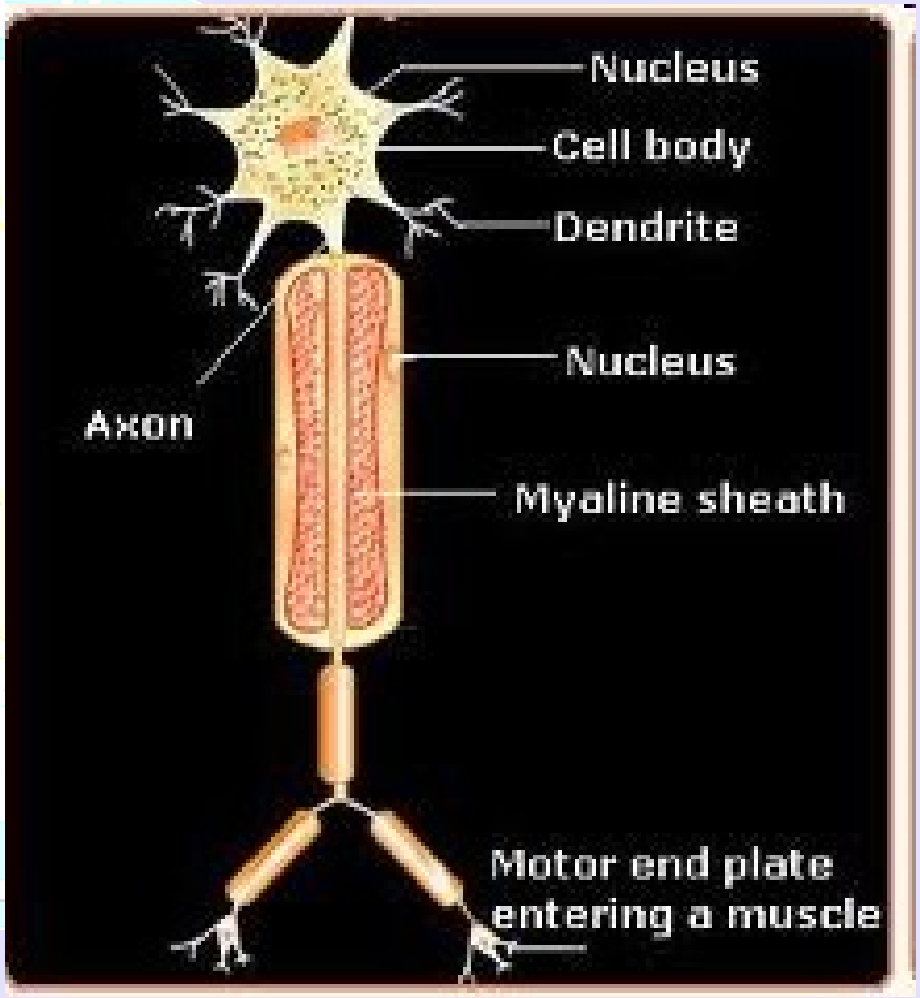


NERVOUS TISSUE: NEURONS

- **Neurons = nerve cells**
  - Cells specialized to transmit messages
  - Major parts of neurons
    - Cell body – nucleus and metabolic center of the cell
    - Processes – fibers that extend from the cell body (dendrites and axons)

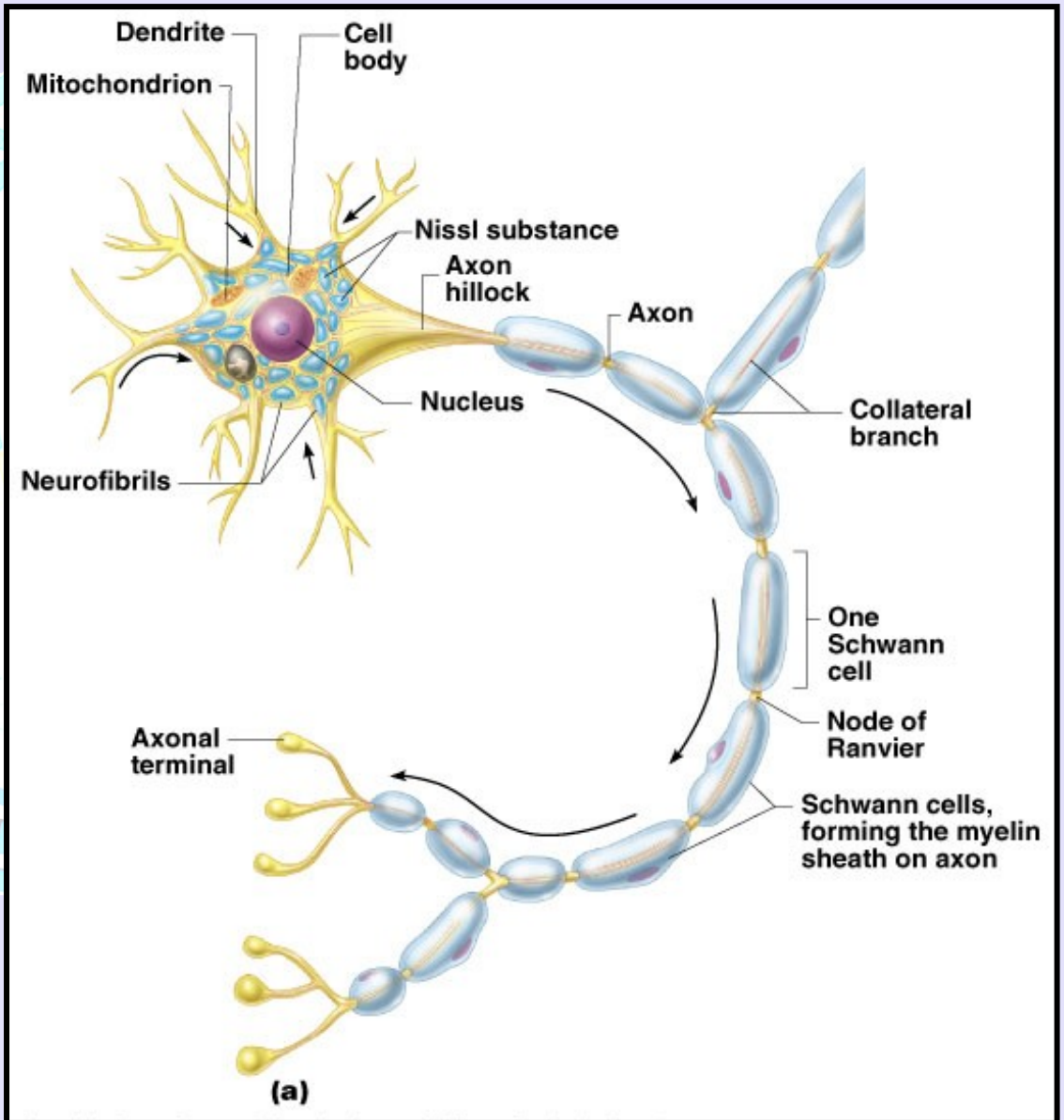
PARTS OF NEURONS

- **Body: (soma)** is the central part of the neuron
- ***Contains***
  - **Nucleus:**
  - **Cytoplasmic organelles**
  - **Inclusions**
  - **Cytoskeletal components**
- **Processes:**
  - **Axon**
  - **Dendrites**



Neuron Anatomy

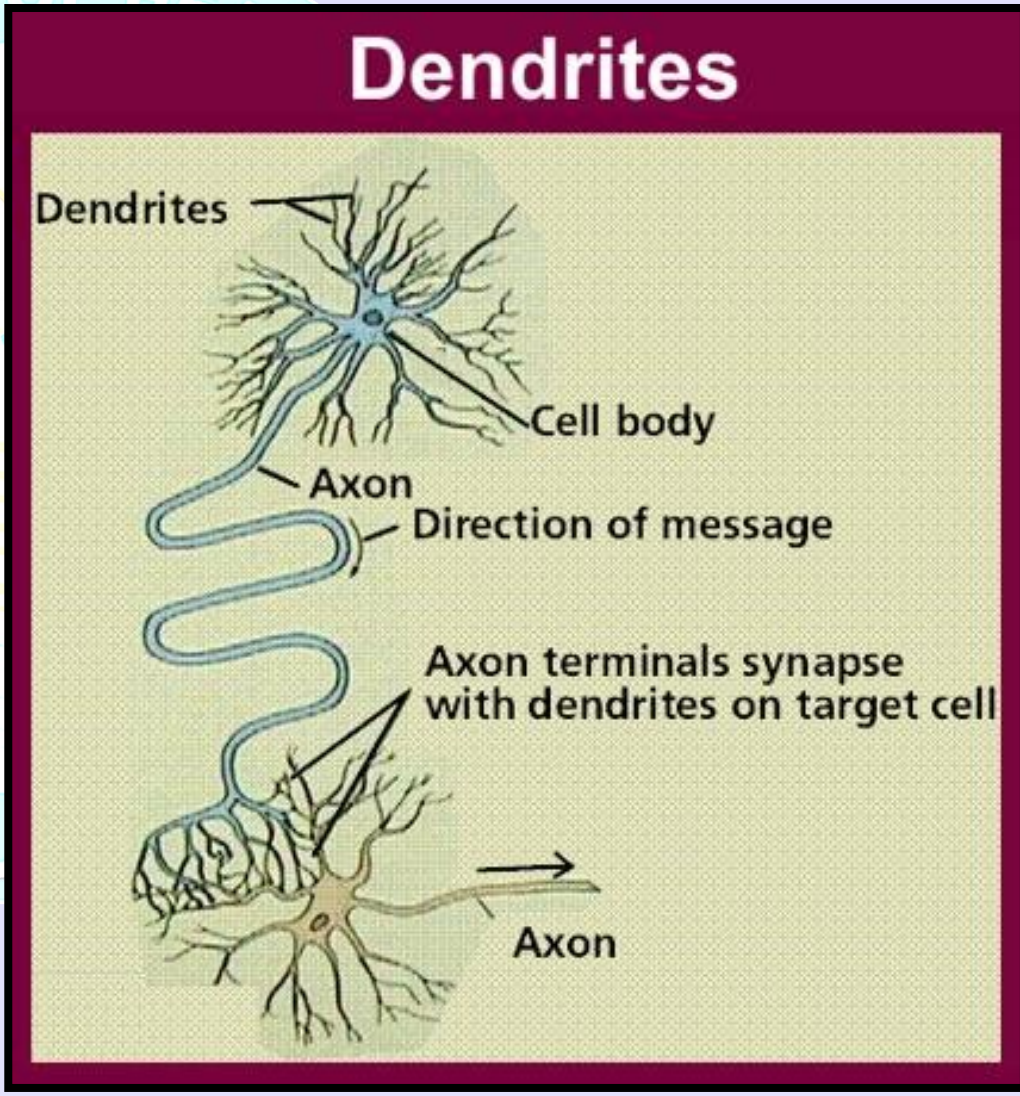
- **Cell body:**
  - Nucleus
  - Large nucleolus



PROCESSES

DENDRITES:

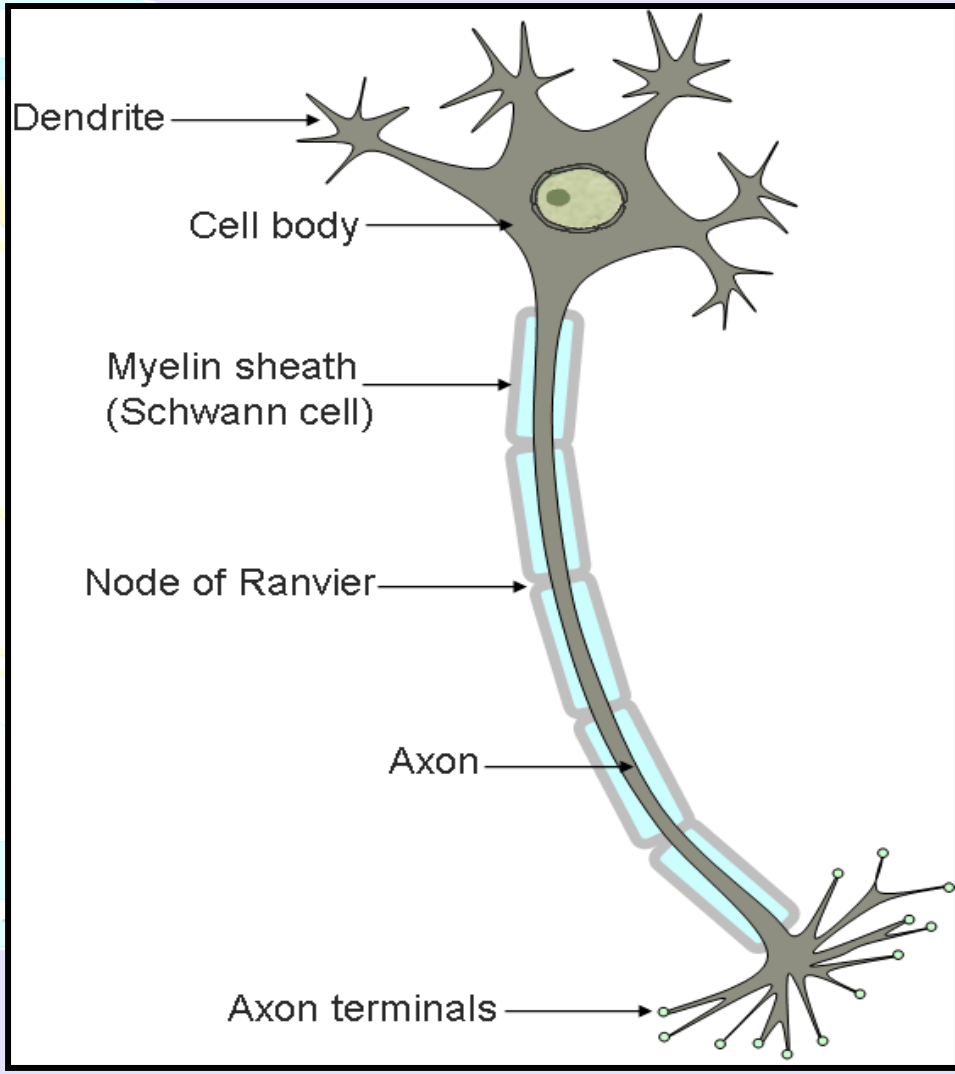
- cellular extensions with many branches
- Input receiving process
- Conveys signals to Soma
- Usually multiple and small in length
- Cytoplasmic organelles similar with Soma
  - only lack golgi complex



PROCESSES

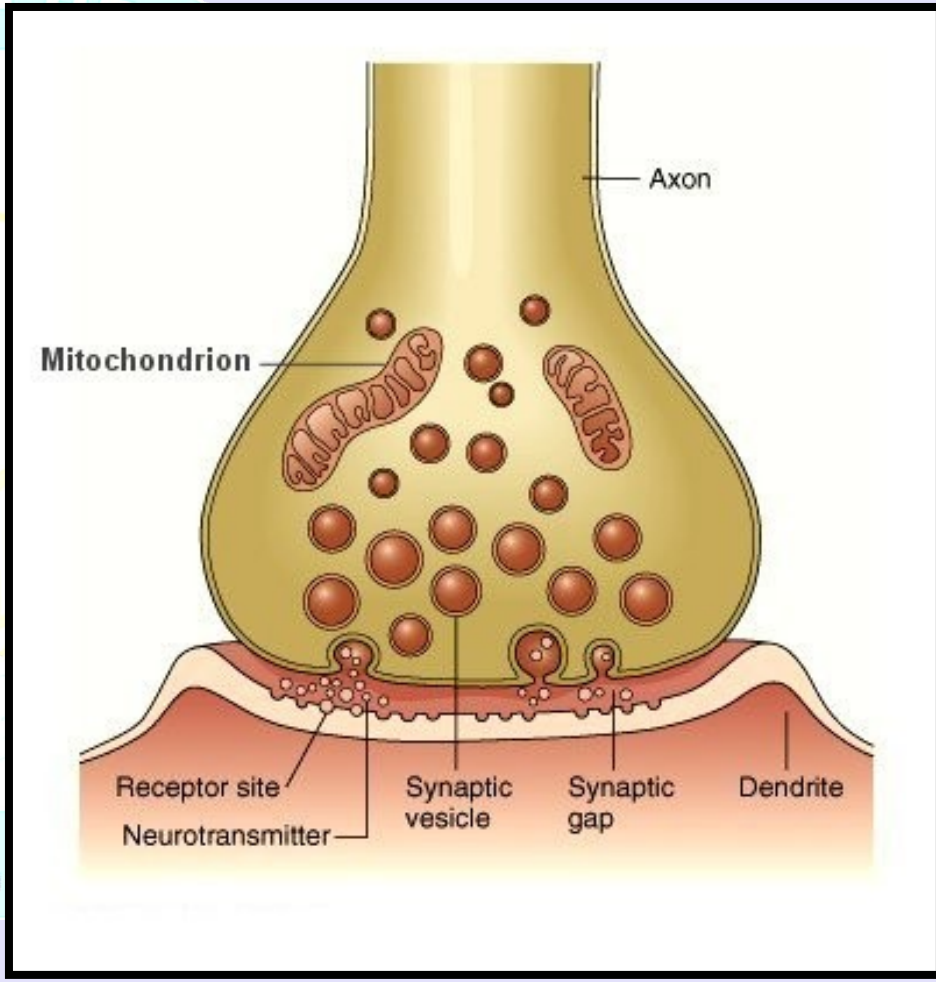
- ***AXON:***

- Output receiving process
- Conduct impulse away from soma
- Different lengths, golgi type-I & type-II
- Single and usually long
- Diameter remains uniform throughout length
- Originates from axon hillock
  - Specialized region of soma



AXON TERMINALS

- **Axon terminals** are distal terminations of the branches of an axon.
- Neurons are interconnected in complex arrangements, and use electrochemical signals and neurotransmitter chemicals to transmit impulses from one neuron to the next.
- axon terminals are separated from neighboring neurons by a small gap called a **synapse,** across which impulses are sent.



REFERENCES

- *Keith L. Moore, Clinically oriented Anatomy,6th edition Page no: 46-49,53-56*

THANK YOU